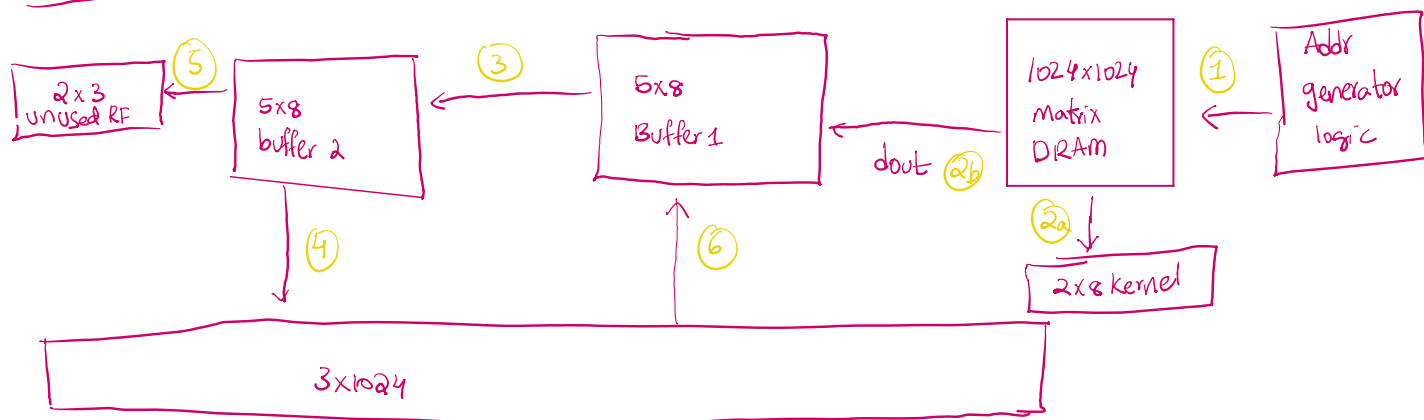
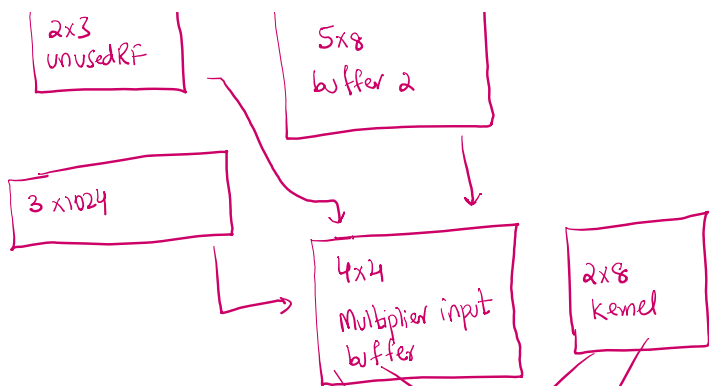


High level data movement with respect to input DRAM



Pipelined MAC

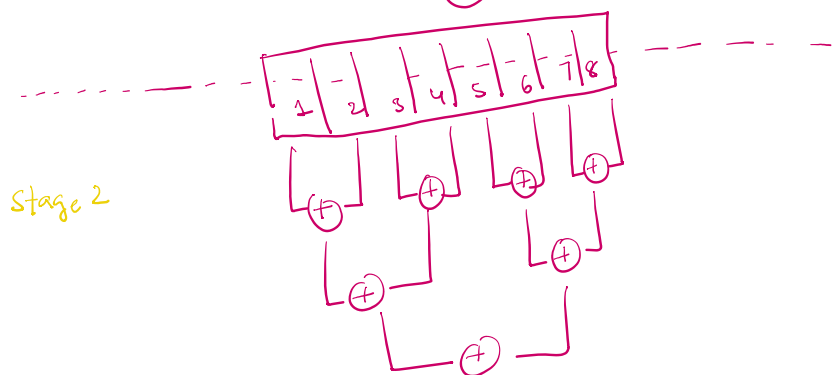




Stage 1

(Repeat for other 14 elements)

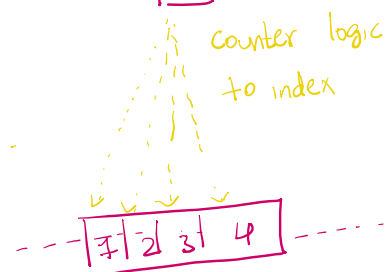
16 multiplier, 8 adder, 8 pipelined Register of 20 bits



Stage 2

1 pipelined Register of 20 bits

Stage 3



counter logic
to index

greater $> 0 \Rightarrow$ stay same
Less $< -4 \Rightarrow$ divide by 4
else $\Rightarrow 0$

Leaky Relu, 4 Register of 20 bits

mux logic to add $1+2+3+4$ (General case)

$1+3$ (Last Row zero padding)

$1+2$ (Last column zero padding)

1 (Last Row zero padding +
Last column zero padding)

Stage 4



Avg pool

mux output $> 4 \Rightarrow$ divide by 4

$L = -4 \Rightarrow \text{divide by } 4$

else $\Rightarrow 0$

